# MobileAgents: Designing and Evaluating a Mobile Multimodal Interface for Controlling Teams of AI Agents On the Go

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## CCS Concepts

• **Human-centered computing** → **Interactive systems and tools**; **Empirical studies in HCI**.

## Keywords

multimodal interaction, agent orchestration, transparency, human-in-the-loop, trust, human–AI interaction

*Word count: 1,450 words (excluding tables and appendices).*

## 1 Introduction: Use Case and Goals

AI Agents are increasingly capable [9, 11, 17]. However, these agents need to be controlled programmatically or through desktop applications. An important use case for these agents is that they can perform tasks while we are away from the workspace. This was demonstrated by OpenClaw [1], an AI agent controlled via messaging apps. This underscores the need for a mobile agent platform that enables frictionless management of agents while on-the-go with adjustable levels of control and governance.

**Multimodal.** We designed a platform that lets users *bring their own AI agents* and manage them via text, voice, and images (Figure 1). Text is sent via a standard conversational interface that supports asynchronous communication with agents. Voice enables hands-free control. Images allow sending pictures from the phone camera.

**Control.** Given that these workflows may execute tasks with varying impact and risks, we prioritize having a highly customizable level of control. Depending on the user's preference, certain tasks are executed completely autonomously, while others require human oversight.

**Stakeholders.** This open-source application is initially targeted at people with a technical background who already utilize agents extensively and wish to extend their reach through mobile devices.

**Persona.** Jessica is a PhD student who runs literature review agents. Jessica wants to monitor progress, redirect agents, and share results with her lab via Slack while commuting.

## 2 Methods: System Design and Implementation

### 2.1 Prototype Overview

MobileAgents is a functional open-source application (FastAPI backend, React frontend, PostgreSQL database, CrewAI orchestration). The system flow is visible in Figure 1. All system prompts are available in Appendix B for reproducibility.

### 2.2 Design Rationale

We prioritize human-centred principles, such as reliability, safety, and transparency [15]. We used Value Sensitive Design (VSD) [5] to inform our design choices.

**V1: Mobility:** "I should be able to reach my agents on the go."

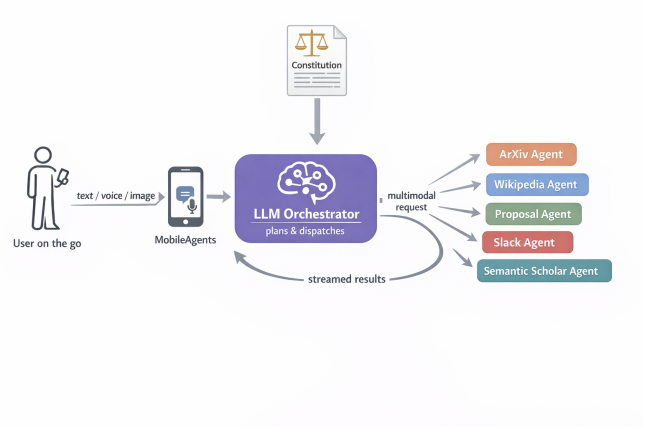



**Figure 1: MobileAgents system overview. Users issue a multimodal request on their mobile device. The LLM orchestrator breaks it down into a plan and assigns tasks to the appropriate agents.**

**V2: Control** "I should understand what the agents are doing and be able to supervise them."

**V3: Ease** "MobileAgents should be simple and intuitive."

**D1:** Support hands-free voice-only mode, images, and text for mobile use cases (Figure 2).

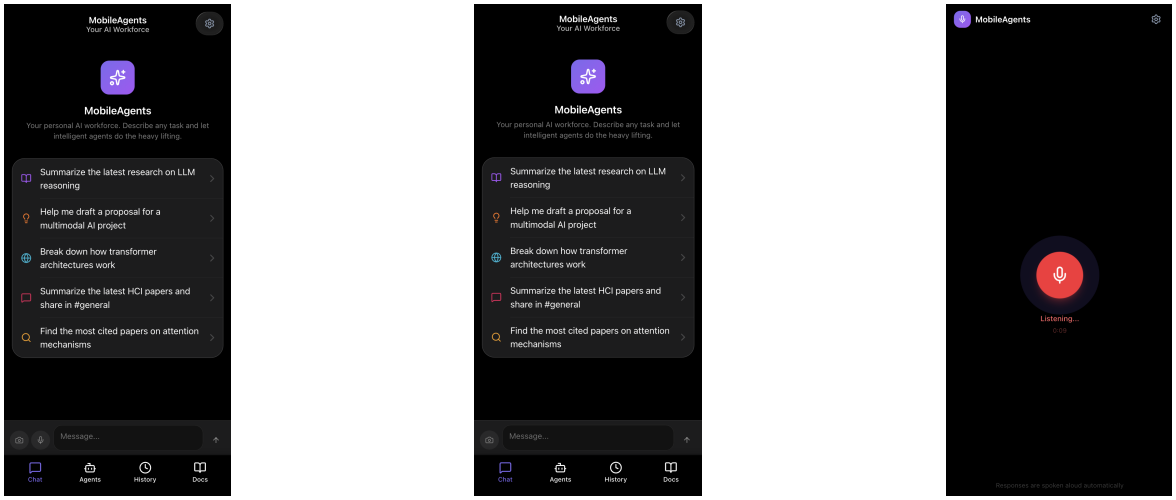
**D2:** Propose a plan for approval before running, render a real-time execution graph, and support customization of orchestration policy.

**D3:** Have simple settings, switches for agents, concise responses, clear visualization, and documentation.

We implement (V1 → D1) through support for text and images, native voice input/output (push-to-talk), and multimodal interaction (Figure 2). Text is the baseline modality. Images capture physical context (e.g., whiteboard) using a camera. The voice-only setting enables hands-free use but does not support plan and transparency controls.

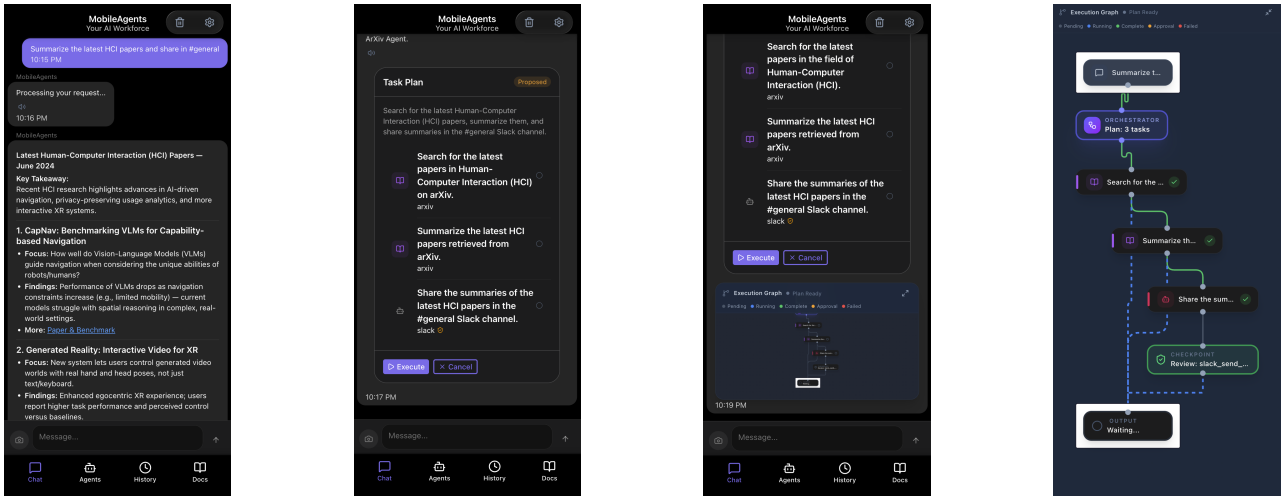
We support control (V2 → D2) through black box, plan, and transparency modes (Figure 3; Table 1) guided by established frameworks on levels of automation [12]. Black box allows full autonomy. Plan mode requests permission before execution. Transparency adds on plan mode with an execution trace graph for complete real-time visibility. Advanced control is available through agent toggles and a natural-language constitution (inspired by Anthropic's Constitutional AI [3]) that guides orchestration policy and stylization.

We address ease (V3 → D3) through simple settings, and the agent management and documentation tabs (Figure 6a, Figure 5b, Figure 4a).




(a) Multimodal mode (b) Text + Images mode (c) Voice-only mode (listening)

**Figure 2: MobileAgents input modes. (a) Multimodal (b) Text + Images. (c) Voice-only**




(a) C1: Black Box (b) C2: Plan Preview (c) C3: Full Transparency (d) Live Tasks Graph

**Figure 3: The three transparency configurations and the C3 graph. (a) Black Box (b) Plan Preview (c) Full Transparency (d) Real-time Execution Graph**

**Table 1: The three transparency configurations compared.**

| Configuration | What the user sees and can do |
|---|---|
| C1: Black Box | Agents auto-execute. |
| C2: Plan Preview | Approve orchestrator's task plan. |
| C3: Transparency | Plan with a real-time graph |

# 3 Methods: Research Question and Evaluation

## 3.1 Research Question

**RQ1:** Is there a perceived usability/control tradeoff between black box, plan, and transparency?

**RQ2:** Does the support of multiple modalities improve the mobile-first experience?

**RQ3:** Are users able to guide orchestration behavior through the advanced agent controls (constitution and agent switches) (Figure 4)?

## 3.2 Study Design

The study comprises three sequential phases.

**Phase 1 — Control (RQ1).** Participants complete the same research task (find recent HCI papers and distribute in Slack) under all control configurations (Table 1)—black box (C1), plan (C2), and transparency (C3)—in counterbalanced order.

**Phase 2 — Modality (RQ2).** Each participant completes a fixed multi-step workflow (find related work and summarize) using all modalities in counterbalanced order: (M1) Multimodal (photograph a printed abstract), (M2) text and images, (M3) voice only (Figure 2). All runs use black-box mode, so modality is the only variable.

**Phase 3 — Orchestration (RQ3).** Participants are asked: "Configure the system so that it never sends Slack messages without showing you a draft first, and remove any agents you do not want included." This is a single exploratory session designed to expose whether the constitution and agent toggles can be located, understood, and utilized (Figure 4).

## 3.3 Participants

We recruited 13 volunteers with experience in LLM-based tools.

## 3.4 Evaluation

In phase 1, participants complete a 7-point Likert survey for *control* and *satisfaction*. In phase 2, they fill the NASA Task Load Index [10] to test cognitive load across modalities, and the Single Ease Question [14] for usability. Participants rate the constitution and toggles on a 3-item perceived controllability scale and shared comments on the user experience. The detailed survey is available in Appendix H.

## 3.5 Metrics

**Table 2: Summary of evaluation metrics by research question.**

| RQ | Metric | Instrument |
|---|---|---|
| RQ1 | Trust, control, satisfaction | Likert (1–7) |
| | Over-/under-informed cases | Qualitative interviews |
| RQ2 | Cognitive load (composite + subscales) | NASA-TLX per modality |
| | Ease of use | Single Ease Question (SEQ) |
| | Modality preference by context | Post-condition survey |
| RQ3 | Perceived controllability | 3-item Likert scale |
| | Discoverability, understanding, ownership | Semi-structured interviews |

# 4 Results

## 4.1 Quantitative Findings

**Table 3: Mean (SD) primary survey scores (1–7 Likert).**

| Metric | C1: Black Box | C2: Plan | C3: Transparency |
|---|---|---|---|
| Trust | 4.3 (1.7) | 5.0 (1.5) | 5.8 (1.0) |
| Control | 3.2 (2.1) | 5.4 (1.3) | 5.7 (1.1) |
| Satisfaction | 4.4 (2.3) | 3.9 (1.1) | 5.8 (1.1) |

Table 3 reveals that Trust and Control monotonically increase from C1 to C3, in alignment with increased interactivity. C3 scores highest (mean 5.8) in satisfaction, whereas C2 performs the worst (3.9), indicating that users preferred either full autonomy or high control. We compare the options using a pairwise Wilcoxon signed-rank test [18] with a Bonferroni correction (Table 4). We do not find significant differences in trust between any pairs. C2 and C3 are significant improvements in control relative to C1. Only C3 has significant results over C2 for satisfaction.

**Table 4: Post-hoc pairwise Wilcoxon signed-rank tests (Bonferroni-corrected, $k = 3$) and Cohen's $d$ for primary RQ1 metrics. $^{*}p_{\text{bon}} < .05$.**

| Metric | Pair | $W$ | $p_{\text{bon}}$ | $d$ |
|---|---|---|---|---|
| Trust | C1-C2 | 16.0 | 0.756 | -0.33 |
| Trust | C1-C3 | 9.5 | 0.120 | -0.69 |
| Trust | C2-C3 | 18.0 | 0.621 | -0.44 |
| Control | C1-C2 | 6.0 | 0.041* | -0.88 |
| Control | C1-C3 | 3.0 | 0.012* | -1.03 |
| Control | C2-C3 | 26.5 | 1.000 | -0.16 |
| Satisfaction | C1-C2 | 29.0 | 1.000 | 0.15 |
| Satisfaction | C1-C3 | 10.0 | 0.234 | -0.58 |
| Satisfaction | C2-C3 | 2.0 | 0.012* | -1.16 |

**Table 5: Mean (SD) SEQ (1–7), NASA-TLX composite and subscales (1–10), and comfort ratings (1–7) per modality.**

| Metric | M1: Multimodal | M2: Text + Image | M3: Voice-only |
|---|---|---|---|
| SEQ (ease, 1–7) | 4.2 (2.0) | 4.5 (2.1) | 5.4 (1.6) |
| TLX composite | 4.2 (0.6) | 4.1 (0.9) | 4.6 (0.6) |
| Mental | 5.5 (2.6) | 5.7 (2.9) | 5.6 (2.6) |
| Physical | 1.8 (0.9) | 1.8 (0.9) | 2.0 (0.8) |
| Temporal | 2.6 (1.4) | 2.8 (1.7) | 5.0 (1.6) |
| Performance | 7.3 (1.5) | 6.6 (1.5) | 5.8 (2.1) |
| Effort | 3.7 (1.9) | 4.2 (2.1) | 5.6 (1.8) |
| Frustration | 4.5 (1.6) | 3.5 (1.5) | 3.4 (2.1) |
| Felt natural (1–7) | 4.2 (1.1) | 4.9 (1.7) | 4.1 (2.0) |
| OK in public (1–7) | 4.0 (1.6) | 3.3 (1.4) | 2.8 (1.3) |

Table 5 reports on modality comparison (RQ2), showing no single modality dominating with statistical significance. That being said, voice scored highest on SEQ (5.4, easiest) but also highest on NASA-TLX composite (4.6, most demanding), and lowest on being natural (4.1) and appropriate in public (2.8). Text + Image scored highest for being natural (4.9), and Multimodal scored best in public (4.0). Overall, the modalities are complementary and are all kept in the settings (Figure 6).

For RQ3, a one-sample Wilcoxon showed controllability ($p = .001$), effectiveness ($p = .016$), and ownership ($p = .036$) were above neutral. Discoverability ($p = .545$) and adoption ($p = .717$) did not differ from neutral. This suggests the constitution is difficult to find and use. The most trusted feature was the execution graph, while the most important feature missing was an undo functionality.

**Table 6: Mean (SD) customisation items (1–7 Likert), RQ3.**

| Item | Mean (SD) |
|---|---|
| Discoverability | 3.8 (1.6) |
| Mental Model | 4.5 (1.8) |
| Effectiveness | 5.2 (1.4) |
| Controllability | 5.7 (1.0) |
| Ownership | 4.8 (1.1) |
| Adoption | 4.2 (1.5) |

## 4.2 Qualitative Assessment

**Intervention is powerful with high observability.** Participants valued the ability to act while observing the real-time execution of tasks. One participant noted: "The graph reveals wrong directions as they begin." This is especially needed in tasks that involve irreversible actions like sending out Slack messages (Figure 5a).

**Multimodal enables mobile.** Participants strongly valued voice for on-the-go use but refused to use it in public spaces, consistent with past research [13]. They reported images as useful for quickly capturing notes. Text remains the standard modality for sensitive tasks.

# 5 Discussion and Future Work

## 5.1 Ethical, Social, and Value-Related Reflection

**Privacy and Security.** We recognize that a centralized manager introduces risks of data storage and prompt injection [8] by malicious agents. Future work should assess the risks and integrate guardrails and safety mechanisms to prevent adversarial attacks by agents in the system.

**Ethics.** This application may promote unhealthy habits and work exploitation using smartphones [4, 6] while away from the workplace. Furthermore, we are concerned about potential cognitive offloading [7, 16] from the abuse of AI tools. Future work should research the impact of MobileAgents on human wellbeing.

**Inclusivity and Access.** MobileAgents is designed for inclusivity and accessibility, supporting multiple languages and themes (Figure 6, Appendix G). We aim to lower the technical barrier in the future by providing an interface that connects external agents through MCP [2] and APIs.

**Limitations.** Our evaluation is preliminary and has many constraints. Our participants are all university students with LLM experience, which does not accurately reflect the entire target population. The single research task used in Phase 1 may not generalize to other agentic workflows, such as code execution or data analysis. Finally, the low statistical power ($N = 13$) is insufficient for conclusive results.

**Conclusion.** We designed a mobile agent management platform balancing automation, control, and usability. Our evaluation explores the tension between control and efficiency through oversight and multimodality. We show that different use cases benefit from different levels of control and that modalities are complementary. Based on this, we release a highly customizable open-source system targeted at technical users and propose future directions grounded in ethics.

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# A Project Structure

```
agentflow/
+-- package.json          # Monorepo scripts
+-- client/
|   \-- src/
|       +-- App.tsx       # Root (auth, theme, RTL)
|       +-- index.css     # Tailwind + animations
|       +-- state/
|       |   \-- store.ts  # Zustand (state + API)
|       +-- types/        # TS interfaces
|       +-- i18n/         # 4 languages (en/zh/ar/bg)
|       \-- components/
|           +-- Chat/     # Chat + markdown + TTS
|           +-- Agents/   # List + detail + toggle
|           +-- Graph/    # ReactFlow DAG + nodes
|           +-- TaskPlan/ # Plan card + approval
|           +-- Camera/   # Image capture + consent
|           +-- Voice/    # Recording + consent
|           +-- Settings/ # Transparency/theme/i18n
|           +-- Execution/# History + step stream
|           \-- Privacy/  # Consent dialog
+-- server/
|   +-- main.py           # FastAPI + CORS + routers
|   +-- config.py         # Pydantic settings
|   +-- auth.py           # Supabase JWT auth
|   +-- routers/
|   |   +-- chat.py       # POST /api/chat
|   |   +-- execute.py    # POST /api/execute (SSE)
|   |   +-- approve.py    # Checkpoint approval
|   |   +-- agents.py     # Agent CRUD
|   |   \-- conversations.py # Conversation CRUD
|   +-- crew/
|   |   +-- orchestrator.py # Planning + graph
|   |   +-- agents.py      # Backstories
|   |   \-- tools.py       # Tool implementations
|   +-- models/
|   |   +-- messages.py   # ChatRequest/Response
|   |   +-- tasks.py      # TaskPlan/TaskStep
|   |   \-- graph.py      # Graph state models
|   \-- services/
|       +-- execution_tracker.py # Dependency exec
|       +-- image_analyzer.py    # GPT-4.1 vision
|       +-- audio_transcriber.py # Whisper
|       +-- agent_store.py       # JSON agent CRUD
|       +-- db.py                # Supabase DB layer
|       \-- supabase_client.py   # Supabase client
```

# B System Prompts

## B.1 Orchestrator Backstory

```
You are a concise research librarian orchestrating
an academic team. Keep all outputs brief — this is
a mobile app with limited screen space.

Break requests into the fewest steps needed.
Prefer parallel execution.
Use uncertainty-aware language when appropriate.

Produce structured JSON plans. Do not add
unnecessary steps.

Available agents:
- arxiv: ArXiv Research Analyst —
    capabilities: arxiv_search, arxiv_summarize
- proposal: Research Proposal Strategist —
    capabilities: generate_proposal,
    outline_methodology
- wikipedia: Background Research Specialist —
    capabilities: wiki_search, wiki_summarize
- slack: Slack Messenger —
    capabilities: slack_send_message
    [requires_approval]
- semantic_scholar: Citation Analyst —
    capabilities: semantic_scholar_search,
    semantic_scholar_cite

Constitution (user-defined guidelines):
<user-editable free text appended here>
```

## B.2 Planning Task Prompt

```
Analyze the following user request and create a
structured execution plan.

User request: {user_message}

[Image analysis: {image_analysis}]
[Audio transcript: {audio_transcript}]
[Conversation history: {history}]

Input modality: {input_modality}
{modality_note}

Available agents:
{agent_descriptions}

Rules:
- Match the user's request to the most appropriate
  agent(s). Agents are NOT limited to research —
  some send messages, perform actions, etc.
- For agents marked [requires_approval], set
  requires_approval=true on that step. For all
  other agents, set requires_approval=false.
- Only return an empty steps list if the request
  truly cannot be handled by ANY available agent
  (e.g. "hello", pure casual chat).
- If steps are independent, leave depends_on empty
  so they can run in parallel.
- If a step needs output from another step, add
  that step's ID to depends_on.
- Always prefer parallel execution when possible.
- Use step IDs like "step_1", "step_2", etc.
- agent_id must be one of: {valid_agent_ids}
- Use exact param names from tool signatures.
  For slack_send_message use params:
  {"channel": "<channel>", "text": "<message>"}
- If image analysis is provided, use that content
  to inform search queries and proposal topics.
- If audio was transcribed, treat the transcript
  as the primary user intent.
- If conversation history is provided, use it to
  resolve ambiguous references.
```

## B.3 Synthesis Prompt

**System prompt:**

```
You are a research orchestrator synthesizing agent
results for a MOBILE interface. Responses are read
on small screens.

Guidelines:
- Be concise: aim for 150-300 words max
- Lead with the key finding or answer in 1-2
  sentences
- Use short bullet points, not long paragraphs
- Use markdown: **bold** for emphasis, short
  headings, compact lists
- Keep relevant URLs but don't list more than 3-5
- Do NOT mention the agents or internal execution
  process
- Use uncertainty-aware language: 'based on
  available results' rather than absolute claims
- End with a one-line disclaimer: '*Results may
  not be exhaustive — verify for critical
  decisions.*'
```

**User message template:**

```
**User request:** {user_message}

**Plan summary:** {plan_summary}

**Agent results:**

### Agent step: {step_description}
{step_result}

### Agent step: {step_description}
{step_result}
...
```

## B.4 Image Analysis Prompt

```
Analyze this image in the context of a research
assistant. Describe what you see and what research
tasks might be relevant (e.g., paper search,
literature review, proposal generation).
Be concise.
```

# C Agent Backstories

```
[ArXiv Agent]
You are an expert research analyst who monitors arXiv
daily. You can search for papers by topic, summarize
their key findings, and identify trends across recent
publications. You excel at distilling complex papers
into clear, actionable summaries.

[Proposal Agent]
You are a seasoned research strategist who has helped
write dozens of successful grant proposals and research
plans. You understand how to identify research gaps,
formulate compelling research questions, and outline
rigorous methodologies. You synthesize information from
literature into coherent proposals.

[Wikipedia Agent]
You are a thorough background researcher who uses
Wikipedia to gather foundational knowledge on topics.
You find relevant articles, extract key concepts and
definitions, and provide well-structured overviews that
help contextualize research work.

[Slack Agent]
You are a communication specialist who sends messages
to Slack channels on behalf of the team. You craft
clear, well-formatted messages and deliver them to the
correct channels. You always confirm which channel to
post to and summarize what was sent.

[Semantic Scholar Agent]
You are a citation analyst who uses Semantic Scholar to
find papers and assess their impact. You search across
all academic literature (not just arXiv), retrieve
citation counts and influential citation metrics, and
help identify the most impactful work in any research
area.
```

# D Tool Implementations

## D.1 SSE Event Protocol

Execution updates are streamed as Server-Sent Events.

# E Data Schemas

## E.1 Request and Response Models

**ChatRequest** (client → server):

```
{
  "message": "string",
  "image_base64": "string | null",
  "audio_base64": "string | null",
  "input_modality": "text | voice | image",
  "conversation_history": [
    {"role": "user|assistant", "content": "..."}
  ],
  "conversation_id": "string | null"
}
```

**ChatResponse** (server → client):

```
{
  "message": "string",
  "plan": TaskPlan | null,
  "graph": ExecutionGraphState | null,
  "image_base64": "string | null"
}
```

## E.2 Task Plan Models

**TaskPlan**:

```
{
  "id": "string",
  "summary": "string",
  "steps": [TaskStep],
  "status": "proposed | executing | completed",
  "created_at": "ISO 8601"
}
```

**TaskStep**:

```
{
  "id": "step_1",
  "agent_id": "arxiv | proposal | wikipedia
              | slack | semantic_scholar",
```

**Table 7: Complete tool registry. Each function is mapped to one or more agents and called directly during execution.**

| Function | Agent | Parameters |
|---|---|---|
| `arxiv_search` | ArXiv | `query`: str, `max_results`: int (default 5) |
| `arxiv_summarize` | ArXiv | `paper_url`: str |
| `generate_proposal` | Proposal | `topic`: str, `context`: str |
| `outline_methodology` | Proposal | `approach`: str, `domain`: str |
| `wiki_search` | Wikipedia | `query`: str |
| `wiki_summarize` | Wikipedia | `title`: str |
| `slack_send_message` | Slack | `channel`: str, `text`: str |
| `semantic_scholar_search` | Sem. Scholar | `query`: str, `max_results`: int (default 5) |
| `semantic_scholar_cite` | Sem. Scholar | `paper_id`: str |

**Table 8: SSE event types streamed during execution.**

| Event type | Payload and semantics |
|---|---|
| `graph_init` | Full `ExecutionGraphState`. Sent once at the start. |
| `node_status` | `nodeId`, `status` (`running` \| `completed` \| `failed` \| `awaiting_approval` \| `approved`), optional `result` and `duration` (ms). |
| `edge_status` | `edgeId`, `status` (`active` \| `completed`). |
| `checkpoint_reached` | `nodeId`, `stepId`. Emitted when an approval-gated step is ready. |
| `execution_complete` | Updated `ExecutionGraphState` and synthesised `summary`. |
| `execution_failed` | `error` message string. |

```
  "action": "tool function name",
  "description": "human-readable description",
  "params": { "key": "value" },
  "status": "pending | running | completed
            | failed | awaiting_approval",
  "result": "string | null",
  "requires_approval": false,
  "depends_on": ["step_id"]
}
```

## E.3 Execution Graph Models

**ExecutionGraphState**:

```
{
  "task_id": "string",
  "nodes": [GraphNode],
  "edges": [GraphEdge],
  "current_node_id": "string | null",
  "status": "planning | executing
            | completed | failed"
}
```

**GraphNode**:

```
{
  "id": "string",
  "type": "input | orchestrator | agent
          | checkpoint | output",
  "data": {
    "label": "string",
    "type": "...",
    "status": "pending | running | completed
              | failed | awaiting_approval
              | approved | skipped",
    "agent_id": "string | null",
    "agent_color": "#hex | null",
    "agent_icon": "LucideIconName | null",
    "result": "string | null",
    "duration": "milliseconds | null",
    "input_modality": "text|voice|image | null",
    "timestamp": "ISO 8601 | null"
  },
  "position": { "x": 0, "y": 0 }
}
```

**GraphEdge**:

```
{
  "id": "e-source-target",
  "source": "node_id",
  "target": "node_id",
  "data": {
    "status": "pending | active | completed",
    "data_preview": "string | null"
  }
}
```

# F Agent Registry Schema

```
{
  "id": "unique_string",
  "name": "Display Name",
  "icon": "LucideIconName",
  "description": "Short description",
  "role": "CrewAI role string",
  "goal": "CrewAI goal string",
  "backstory": "Full system prompt text",
  "capabilities": ["tool_1", "tool_2"],
  "enabled": true,
  "requiresApproval": false,
  "color": "#hex",
  "isOrchestrator": false,
  "constitution": ""
}
```

# G CSS Animations and Theming

## G.1 Theme Variables

# H Survey Instrument

All Likert items use a 7-point scale (1 = Strongly Disagree, 7 = Strongly Agree) unless otherwise noted.

**Table 9: Theme colour tokens (selected).**

| Token | Light | Dark |
|---|---|---|
| `-surface-0` | `#ffffff` | `#000000` |
| `-surface-1` | `#f2f2f7` | `#1c1c1d` |
| `-surface-2` | `#e5e5ea` | `#121212` |
| `-text-primary` | `#000000` | `#ffffff` |
| `-text-secondary` | `#3c3c43` | `#ffffffcc` |

## Part 0: Background (before study begins)

(1) Age range. *(18–21 / 22–25 / 26–30 / 31–40 / 40+)*
(2) Gender. *(Male / Female / Non-binary / Prefer not to say)*
(3) How often do you use AI assistants (e.g., ChatGPT, Copilot)? *(Daily / Several times a week / Weekly / Monthly / Rarely or Never)*
(4) Have you used a multi-agent AI system before? *(Yes / No / I'm not sure)*
(5) Which AI tools do you use regularly? *(Select all: ChatGPT, Copilot, Gemini, Claude, Cursor, Perplexity, Other)*

## Part 1: After each transparency condition (C1, C2, C3) — RQ1

Completed three times, once per configuration. Items 1–3 are primary metrics; items 4–7 are supplementary.

(1) I trusted the system's output enough to act on it. *(Trust)*
(2) I felt in control of what the system did on my behalf. *(Perceived Control)*
(3) I am satisfied with this interaction. *(Satisfaction)*
(4) The system made it clear what it was going to do before doing it. *(Expectation Setting)*
(5) It was easy to correct or stop the system when needed. *(Repairability)*
(6) I understood why the system made the decisions it did. *(Explainability)*
(7) I felt comfortable with the information I shared during this task. *(Privacy Comfort)*
– Any comments about this mode? *(open text, optional)*

After all three conditions:

8. Which transparency mode would you prefer for everyday use? *(C1 Black Box / C2 Plan Preview / C3 Full Transparency)*
9. Which mode would you prefer for high-stakes tasks (e.g., sending a Slack message on your behalf)? *(C1 / C2 / C3)*
10. Why did you prefer that mode? *(open text, optional)*

## Part 2: After each modality condition (M1, M2, M3) — RQ2

Completed three times, once per modality (text, image, voice). The NASA-TLX uses a 1–10 scale (Very Low–Very High); Performance is reversed (1 = Perfect, 10 = Failure).

**Single Ease Question (SEQ):**

(1) Overall, how would you rate the difficulty of this task? *(1 = Very Difficult, 7 = Very Easy)*

**NASA Task Load Index [10]:**

2. Mental Demand — How much mental and perceptual activity was required? *(1–10)*
3. Physical Demand — How much physical activity was required? *(1–10)*
4. Temporal Demand — How much time pressure did you feel? *(1–10)*
5. Performance — How successful were you in accomplishing the task? *(1 = Perfect, 10 = Failure)*
6. Effort — How hard did you have to work to accomplish the task? *(1–10)*
7. Frustration — How irritated, stressed, or annoyed did you feel? *(1–10)*

**Modality comfort (7-point Likert):**

8. Using this modality felt natural for this type of task. *(Modality Comfort)*
9. I would use this modality in a public or shared space. *(Context Appropriateness)*
– In what situations would you choose or avoid this modality? *(open text, optional)*

After all three modality conditions:

10. Which input modality would you use most often on a mobile device? *(Text / Voice / Image / It depends)*
11. Which modalities felt well-suited to on-the-go use? *(Select all: Text / Voice / Image)*
– Was there a modality that surprised you – positively or negatively? *(open text, optional)*

## Part 3: After the customisation task (Phase 3) — RQ3

(1) I was able to find the constitution editor without help. *(Discoverability)*
(2) I understood what the constitution field does. *(Mental Model)*
(3) Editing the constitution changed the system's behaviour as I expected. *(Effectiveness)*
(4) The agent on/off toggles were easy to understand and use. *(Controllability)*
(5) After customising the system, I felt more in control. *(Perceived Ownership)*
(6) I would use the constitution and agent controls regularly. *(Adoption Intent)*
7. Which type of change did you make to the constitution? *(Constraint / Style / Routing / Did not edit / Other)*
– Describe what you changed and whether the system behaved as expected. *(open text)*

## Part 4: Overall post-study feedback

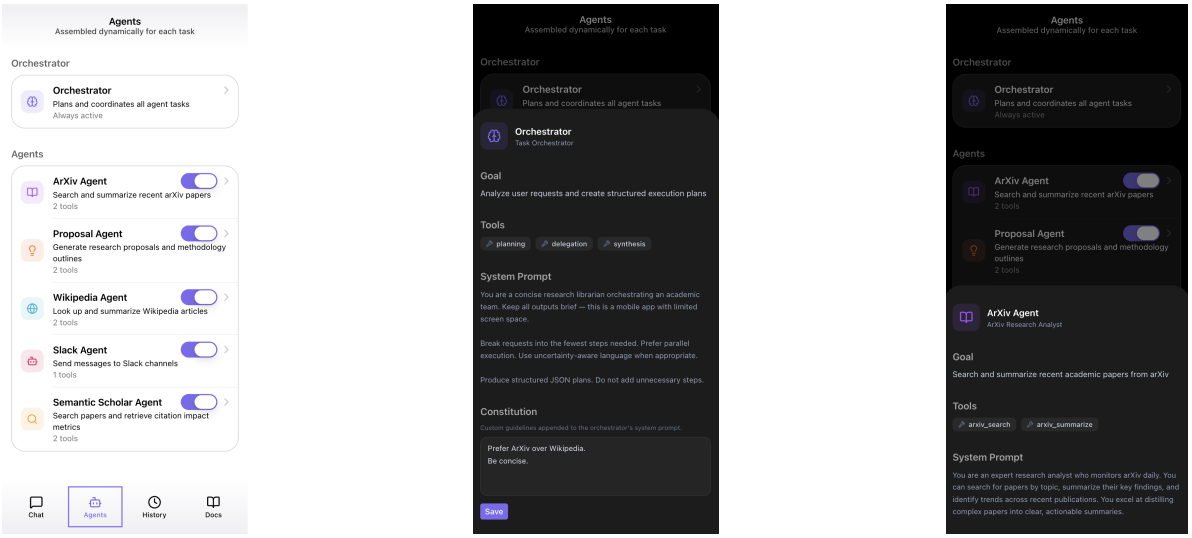
(1) Overall, I would trust this system to act on my behalf on a mobile device. *(Overall Trust)*
(2) I felt this system respected my privacy. *(Privacy)*
(3) I would use this application in my day-to-day work. *(Adoption Intent)*
4. Which single feature contributed most to your trust? *(Plan Preview / Checkpoint approval / Live DAG / Agent on/off / Constitution / Multimodal input)*

5. Which single feature caused the most friction or concern? *(Plan approval steps / Live graph overwhelming / Voice risky in shared spaces / Camera awkward / Agent controls hard to find / Constitution unclear)*
   - What is the single most important improvement you would make? *(open text)*
   - Any other comments or suggestions? *(open text)*

# I Semi-Structured Interview Questions

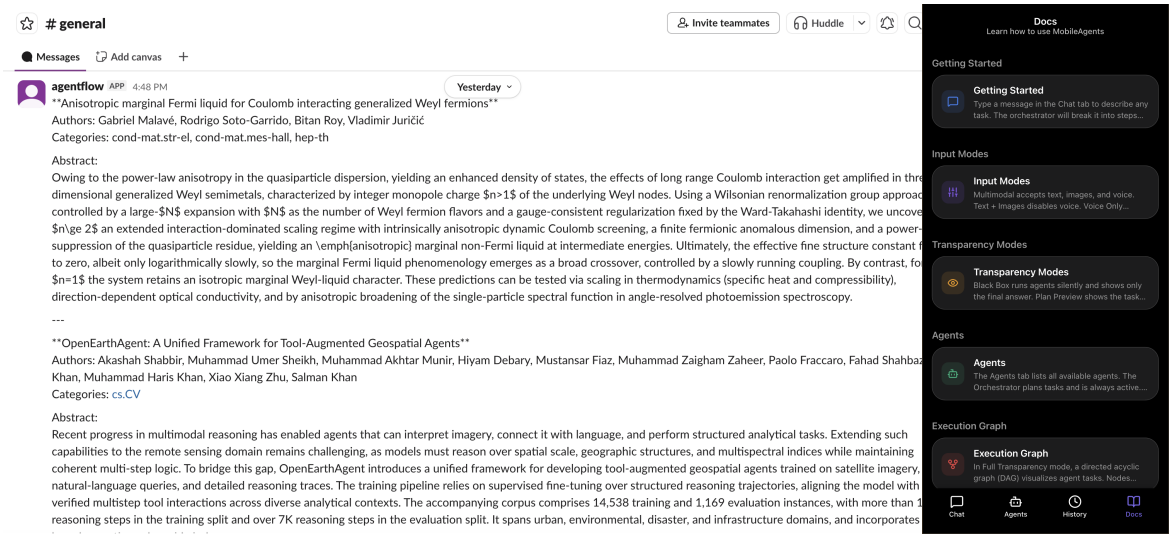
After completing all three configurations, participants were asked:

(1) You used three different versions of the system. Can you describe how each felt different?
(2) Which version would you prefer for everyday use? Why?
(3) Was there a moment where you particularly trusted—or distrusted—the system? What caused that?
(4) The system showed you a plan before executing. Did that change how you interacted with it? How?
(5) For the version with the live graph, was seeing the execution progress helpful or distracting?
(6) The system required approval before sending a Slack message. How did that approval step feel? Too cautious? About right?
(7) How did you decide whether to use text, voice, or camera input? Were there situations where you avoided a modality? Why?
(8) Did any of the system's responses feel overconfident or uncertain? How did that affect your trust?
(9) If this system stored your conversation history across sessions, how would you feel about that? What controls would you want?
(10) Can you think of a situation where a system like this could cause harm—to you or someone else?
(11) What is the single most important change you would make to improve this system?



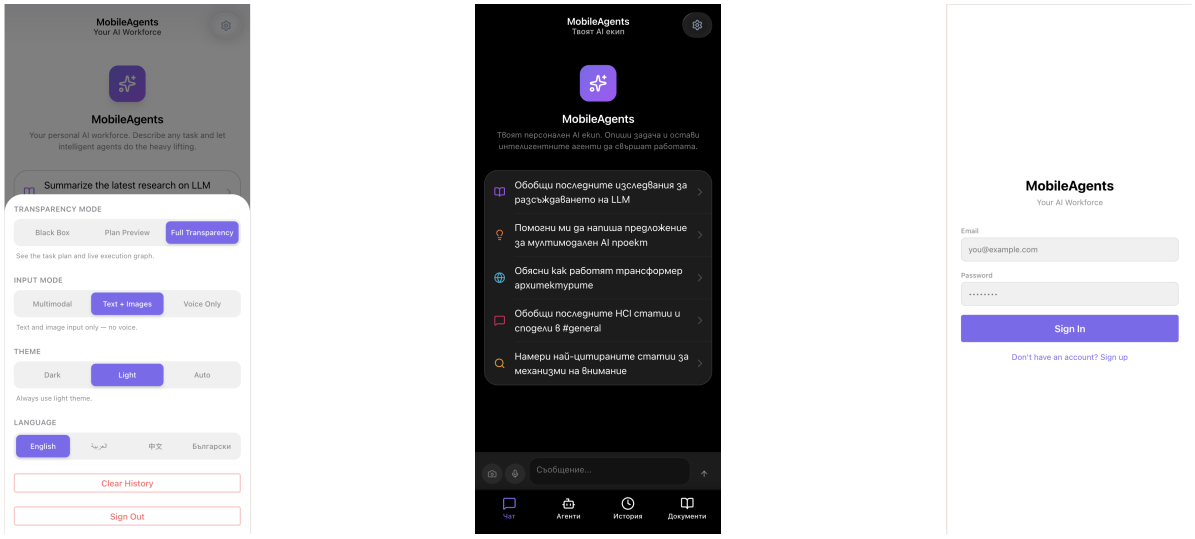

(a) Agents tab (light theme) (b) Orchestrator detail with constitution editor (c) Agent card (ArXiv agent)

**Figure 4: Agent management screens. (a) The Agents tab lists the orchestrator and all registered worker agents with enable/disable toggles. (b) Tapping the orchestrator opens its detail sheet, showing goal, tools, system prompt, and the editable constitution field. (c) A worker agent card shows its goal, tool names, and full system prompt backstory.**




(a) Slack output (#general channel) (b) In-app documentation tab

**Figure 5: Slack message posted to #general by the Slack agent. (b) The Docs tab.**




(a) Settings panel (b) Language localisation (c) Login screen

**Figure 6: Settings, localisation, and login screens. (a) Settings exposes all controls, modalities, themes, and languages (EN/AR/ZH/BG). (b) Language localisation. (c) Login screen with email and password authentication.**